Chapter 7 : Other Regression Techniques

Ordinary Least Squares (OLS)

Recall the linear regression equation

$$E(Y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k$$

We have estimated the parameters β_0, β_1, β_2, $\ldots$, β_k by minimizing the sum of squared residuals

$$\begin{aligned} \text{SSE} &= \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 \\ &= \sum_{i=1}^{n} \left[y_i - \left(\hat{\beta}_0 + \hat{\beta}_1 x_{i,1} + \hat{\beta}_2 x_{i,2} + \cdots + \hat{\beta}_k x_{i,k} \right) \right]^2 . \end{aligned}$$

Sometimes we want to give some observations more weight than others.

Weighted Least Squares (WLS)

We achieve this by minimizing a *weighted* sum of squares:

$$\begin{aligned} \text{WSSE} &= \sum_{i=1}^{n} w_i \, (y_i - \hat{y}_i)^2 \\ &= \sum_{i=1}^{n} w_i \left[y_i - \left(\hat{\beta}_0 + \hat{\beta}_1 x_{i,1} + \hat{\beta}_2 x_{i,2} + \cdots + \hat{\beta}_k x_{i,k} \right) \right]^2 \end{aligned}$$

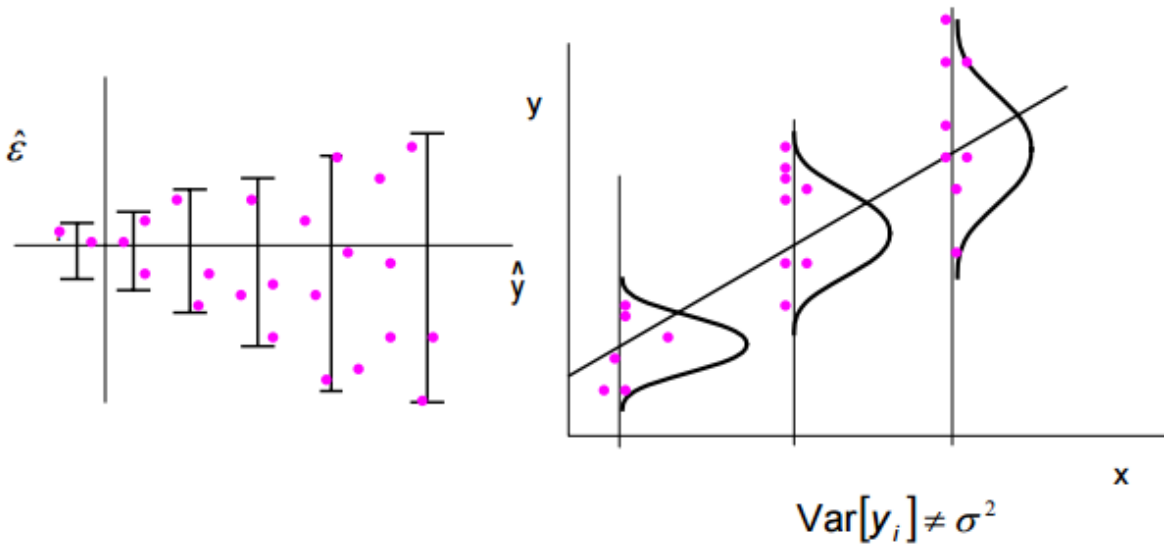
The resulting $\hat{\beta}$s are called *weighted least squares* (WLS) estimates, and the *WLS residuals* are

$$\sqrt{w_i}(y_i - \hat{y}_i).$$

Why use weights?

Suppose that the variance is not constant:

$$\text{var}(Y_i) = \sigma_i^2.$$



If we use weights

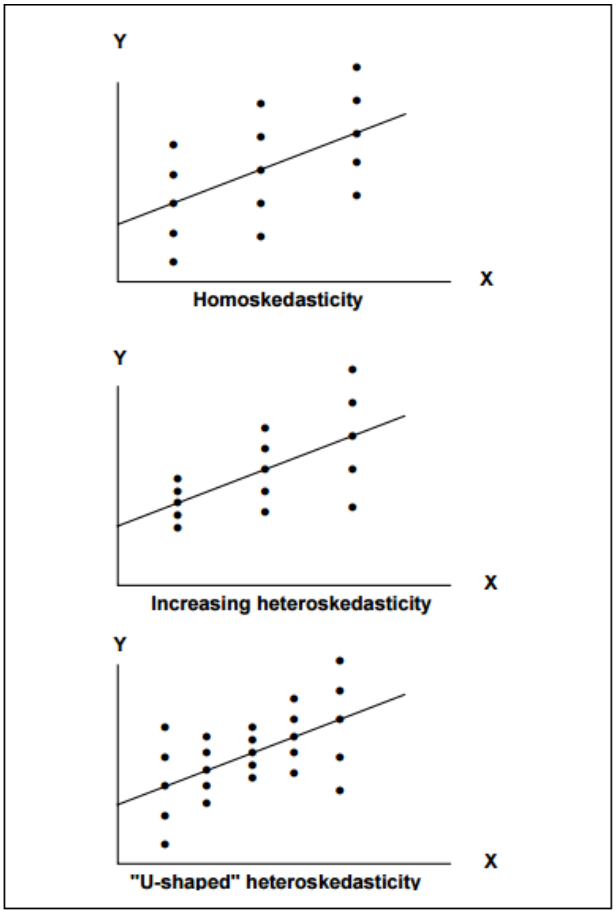
$$w_i \propto \frac{1}{\sigma_i^2},$$

the WLS estimates have smaller standard errors than the ordinary least squares (OLS) estimates.

In fact, using weights proportional to $1/\sigma_i^2$ is optimal: no other weights give smaller standard errors.

When you specify weights, regression software calculates standard errors on the assumption that they are proportional to $1/\sigma_i^2$.

- The main advantage - handle regression situations in which the data points are of varying quality (a solution to Heteroskedasticity). It's well suited to extracting maximum information from small data sets.
- The biggest disadvantage - the method is based on the assumption that the weights are known exactly. Sensitivity to outlier is a problem. An outlier given an inappropriate weight could dramatically skew your results.



Theory

Weighted Least Squares in Simple Regression

* Suppose that we have the following model

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i \qquad i = 1, \ldots, n$$

where $\varepsilon_i \sim \mathrm{N}(0, \sigma^2/w_i)$ for known constants $w_1, \ldots, w_n$.

* The weighted least squares estimates of β_0 and β_1 minimize the quantity

$$S_w(\beta_0, \beta_1) = \sum_{i=1}^{n} w_i (y_i - \beta_0 - \beta_1 x_i)^2$$

* Note that in this weighted sum of squares, the weights are inversely proportional to the corresponding variances; points with low variance will be given higher weights and points with higher variance are given lower weights.

* The weighted least squares estimates are then given as

$$\begin{aligned}\hat{\beta}_0 &= \overline{y}_w - \hat{\beta}_1 \overline{x}_w \\ \hat{\beta}_1 &= \frac{\sum w_i(x_i - \overline{x}_w)(y_i - \overline{y}_w)}{\sum w_i(x_i - \overline{x}_w)^2}\end{aligned}$$

where $\overline{x}_w$ and $\overline{y}_w$ are the weighted means

$$\overline{x}_w = \frac{\sum w_i x_i}{\sum w_i} \qquad \overline{y}_w = \frac{\sum w_i y_i}{\sum w_i}.$$

* Some algebra shows that the weighted least squares estimates are still unbiased.

* Furthermore we can find their variances

$$\mathrm{Var}(\hat{\beta}_1) = \frac{\sigma^2}{\sum w_i(x_i - \overline{x}_w)^2}$$

$$\mathrm{Var}(\hat{\beta}_0) = \left[\frac{1}{\sum w_i} + \frac{\overline{x}_w^2}{\sum w_i(x_i - \overline{x}_w)^2}\right]\sigma^2$$

* Since the estimates can be written in terms of normal random variables, the sampling distributions are still normal.

* The weighted error mean square $S_w(\hat{\beta}_0, \hat{\beta}_1)/(n-2)$ also gives us an unbiased estimator of σ^2 so we can derive t tests for the parameters etc.

General Weighted Least Squares Solution

* Let $\boldsymbol{W}$ be a diagonal matrix with diagonal elements equal to $w_1, \ldots, w_n$.

* The the Weighted Residual Sum of Squares is defined by

$$S_w(\boldsymbol{\beta}) = \sum_{i=1}^{n} w_i(y_i - \boldsymbol{x}_i^t\boldsymbol{\beta})^2 = (\boldsymbol{Y} - \boldsymbol{X}\boldsymbol{\beta})^t\boldsymbol{W}(\boldsymbol{Y} - \boldsymbol{X}\boldsymbol{\beta}).$$

* Weighted least squares finds estimates of $\boldsymbol{\beta}$ by minimizing the weighted sum of squares.

* The general solution to this is

$$\hat{\boldsymbol{\beta}} = (\boldsymbol{X}^t\boldsymbol{W}\boldsymbol{X})^{-1}\boldsymbol{X}^t\boldsymbol{W}\boldsymbol{Y}.$$

Regression Robust

Resistant Regression Methods

- Resistant methods are good at dealing with data where we expect there to be a certain number of "bad" observations that we want to have no weight in the analysis.
- A few types:
  - Least Trimmed Squares (`LTS`) : More efficient compared to LMS.
  - Least Quantile of Squares(`LQS`)
  - Least Median of Squares (`LMS`) : Not recommended for small samples
  - S-estimation (`S`) : More efficient than LTS and LMS when data is normal.
    It is based on residual scale of M-estimation.
  - M-estimation (`M`) : M estimation is an estimation of the maximum likelihood type.
  - MM-estimation (`MM`): It uses S estimation which minimizes the scale of the residual from M estimation and then proceed with M estimation.

In R:

To use **resistant regression**: function `lqs()` # default LTS method

```
## Default S3 method:
lqs(x, y, intercept = TRUE, method = c("lts", "lqs", "lms", "S"),
    quantile, control = lqs.control(...), k0 = 1.548, seed, ...)
```

To use **robust regression** : function `rlm()`

Description

Fit a linear model by robust regression using an M estimator.

Usage

```
rlm(x, ...)

## Default S3 method:
rlm(x, y, weights, ..., w = rep(1, nrow(x)),
    init = "ls", psi = psi.huber,
    scale.est = c("MAD", "Huber", "proposal 2"), k2 = 1.345,
    method = c("M", "MM"), wt.method = c("inv.var", "case"),
    maxit = 20, acc = 1e-4, test.vec = "resid", lqs.control = NULL)
```

Example 1:

```
> lqs(y1 ~ x1, data = anscombe)
Call:
lqs.formula(formula = y1 ~ x1, data = anscombe)

Coefficients:
(Intercept)          x1
     2.6678      0.5233

Scale estimates 0.6557 0.9879
```

You may try the following to see R-output results:

```
lqs(y1~x1,method ="lts")
lqs(y1~x1,method ="lqs")
lqs(y1~x1,method ="lms")
lqs(y1~x1,method ="S")
```

Fitting Linear Models (`lm`)

```
> lm(y1 ~ x1)

Call:
lm(formula = y1 ~ x1)

Coefficients:
(Intercept)           x1
     3.0001       0.5001
```

`? loess`

- It is a way to fit a scatter plot with a smooth fitted line (local polynomial regression fitting)
- It is also known as **moving regression**
- The polynomial is fitted using weighted least squares, giving more weight to points near the point whose response is being estimated and less weight to points further away.